

H4 SF2525

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Preliminaries

0.1

In order to show that

$$X_t = x_0 + at + bW_t$$

with x_0 initial value, solves the stochastic differential equation:

$$dX_t = adt + bdW_t$$

we can simply take the derivative with respect to time and then multiplying both sides of the equation by dt :

$$\begin{aligned}\frac{dX_t}{dt} &= a + b\frac{dW_t}{dt} \\ \implies dX_t &= adt + bdW_t\end{aligned}$$

0.2

Here we are computing the likelihood function $p(x|a, b)$ as the product of the probability densities of increments Δx . Since, the increments are independent and identically distributed, this means that $\Delta x \sim \mathcal{N}(a\Delta t_n, b^2\Delta t_n)$

Let us prove it:

$$\Delta x_n = x_n - x_{n-1} = a(t_n - t_{n-1}) + b(W_n - W_{n-1}) = a\Delta t_n + b\Delta W_n$$

and remembering that $\Delta W_n \sim \mathcal{N}(0, \Delta t_n)$, then from properties of linear transformations, we have that

$$\Delta x \sim \mathcal{N}(a\Delta t_n, b^2\Delta t_n)$$

0.3

See code in attached file

Frequentist Parameter Estimation

1.1

From previous section we know that the likelihood is

$$p(x|a, b) = \prod_{n=1}^N \frac{1}{b\sqrt{2\pi\Delta t_n}} \exp\left(-\left(\frac{\Delta x_n - a\Delta t_n}{b\sqrt{2\Delta t_n}}\right)^2\right)$$

starting from this, using the properties of the exponential function and logarithmic function, we can derive $-\log(p(x|a, b))$:

$$\begin{aligned}
-\log(p(x|a, b)) &= -\log\left(\prod_{n=1}^N \frac{1}{b\sqrt{2\pi\Delta t_n}} \exp\left(-\left(\frac{\Delta x_n - a\Delta t_n}{b\sqrt{2\pi\Delta t_n}}\right)^2\right)\right) = \\
&= -\sum_{n=1}^N \log\left(\frac{1}{b\sqrt{2\pi\Delta t_n}}\right) + \log\left(\exp\left(\frac{-(\Delta x_n - a\Delta t_n)^2}{2b^2\Delta t_n}\right)\right) = \\
&= \sum_{n=1}^N \log(b\sqrt{2\pi\Delta t_n}) + \frac{(\Delta x_n - a\Delta t_n)^2}{2b^2\Delta t_n} = \\
&= \sum_{n=1}^N \frac{(\Delta x_n - a\Delta t_n)^2}{2b^2\Delta t_n} + \sum_{n=1}^N \log(b\sqrt{2\pi\Delta t_n})
\end{aligned}$$

We can analyze the sum of the logarithmic term and decompose it in the following way:

$$\begin{aligned}
\sum_{n=1}^N \log(b\sqrt{2\pi\Delta t_n}) &= \sum_{n=1}^N \log(b) + \frac{1}{2} \log(2\pi\Delta t_n) = \\
&= N \log(b) + \frac{N \log(2\pi)}{2} + \frac{1}{2} \sum_{n=1}^N \log(\Delta t_n)
\end{aligned}$$

Thus, putting all together we proved:

$$-\log(p(x|a, b)) = N \log(b) + \frac{N \log(2\pi)}{2} + \frac{1}{2} \sum_{n=1}^N \log(\Delta t_n) + \sum_{n=1}^N \frac{(\Delta x_n - a\Delta t_n)^2}{2b^2\Delta t_n}$$

1.2

Now we aim to derive an expression for the minimizer (a_{ML}, b_{ML}) of the function $-\log(p(x|a, b))$.

We'll proceed deriving the function (first time with respect to a and then with respect to b) and setting these derivatives at zero (checking the corresponding second derivatives in those points have positive values to ensure they are actually minimizer).

$$\begin{aligned}
\frac{\partial}{\partial a} [-\log(x | a, b)] &= \frac{1}{b^2} (a \sum_{n=1}^N \Delta t_n - \sum_{n=1}^N \Delta x_n) = 0 \\
\Rightarrow a_{ML} &= \frac{\sum_{n=1}^N \Delta x_n}{\sum_{n=1}^N \Delta t_n} = \frac{x_N - x_0}{t_N - t_0}
\end{aligned}$$

Similarly for b_{ML} :

$$\begin{aligned}
\frac{\partial}{\partial b} [-\log(x | a, b)] &= \frac{N}{b} - \frac{1}{b^3} \sum_{n=1}^N \frac{(\Delta x_n - a\Delta t_n)^2}{\Delta t_n} = 0 \\
\Rightarrow b_{ML}^2 &= \frac{1}{N} \sum_{n=1}^N \frac{(\Delta x_n - a_{ML}\Delta t_n)^2}{\Delta t_n}
\end{aligned}$$

1.3

Now we want to derive the expression for

$$p(X_{N+1}|X, a_{ML}, b_{ML})$$

where $X_{N+1} = X(t_{N+1}) = X(1.5)$ First, considering the Markov property of the stochastic process, we can state that:

$$p(X_{N+1}|X, a_{ML}, b_{ML}) = p(X_{N+1}|x_N, a_{ML}, b_{ML})$$

From the SDE solution, with $a = a_{ML}$ and $b = b_{ML}$ we have:

$$X_{N+1} = x_N + a_{ML}\Delta t_{N+1} + b_{ML}\Delta W_{N+1}$$

and using that $\Delta W_{N+1} \sim \mathcal{N}(0, \Delta t_{N+1})$ and the properties of linear transformations, we get:

$$(X_{N+1}|X) \sim \mathcal{N}(x_N + a_{ML}\Delta t_{N+1}, b_{ML}^2\Delta t_{N+1})$$

Particularly, in our case with $t_{N+1} = 1.5$ and $t_N = 1$ we have:

$$(X_{N+1}|X) \sim \mathcal{N}(x_N + 0.5 * a_{ML}, 0.5 * b_{ML}^2)$$

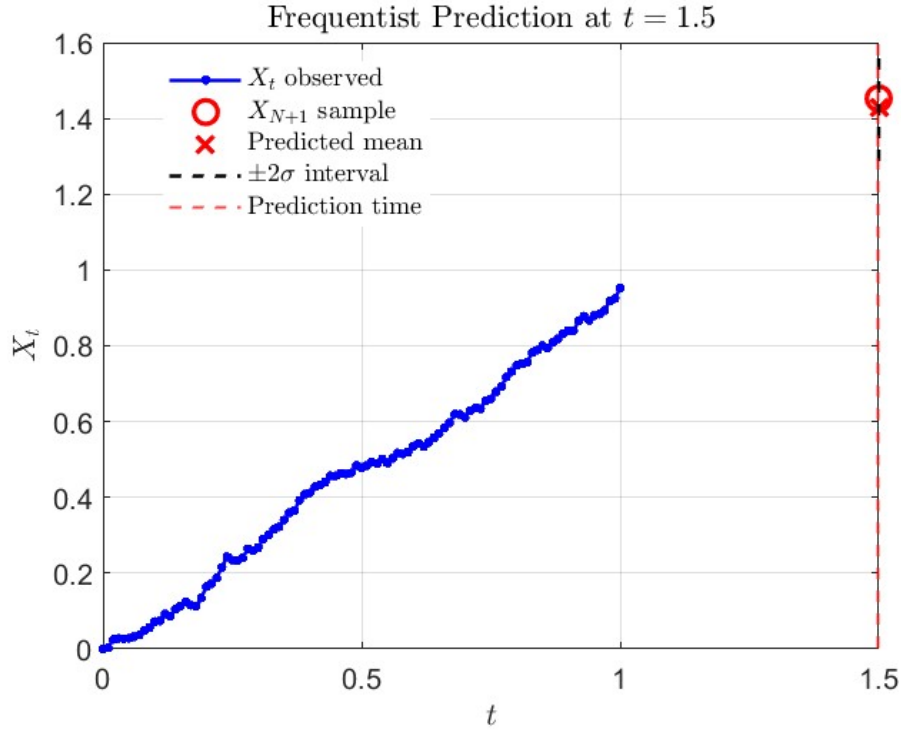


Figure 1: X_{N+1} prediction frequentist approach

Bayesian Parameter Estimation

2.1

The goal here is to show that the probability density distribution of A given X and B^2 , i.e. $p(A|X, B^2)$, is gaussian and derive an expression for its mean value and variance.

Using Bayes' rule we can write:

$$p(A|X, B^2) = \frac{p(X|A, B^2)p(A)}{p(X)} \propto p(X|A, B^2)p(A)$$

(notice that in this case $p(X)$ is a constant, since the sample X is given, so in order to simplify computations, we are going to consider a proportionality relationship between the left hand side and right hand side terms). Before proceeding, we can prove the inequality:

$$p(A|X, B^2) = \frac{p(A, (X, B^2))}{p(X, B^2)}$$

and

$$p(X|A, B^2) = \frac{p(X, (A, B^2))}{p(A, B^2)} \implies p(X, (A, B^2)) = p(X|A, B^2)p(A, B^2)$$

Since $p(A, (X, B^2)) = p(X, (A, B^2))$, we can substitute and using independency of variables we obtain:

$$p(A|X, B^2) = \frac{p(X|A, B^2)p(A, B^2)}{p(X, B^2)} = \frac{p(X|A, B^2)p(A)}{p(X)}$$

Now we want to try to write $p(X|A, B^2)p(A)$ in such a way that we can recognize the gaussian form:

$$\begin{aligned} p(X|A, B^2)p(A) &= \prod_{n=1}^N \frac{1}{b\sqrt{2\pi}\Delta t_n} \exp\left(-\left(\frac{\Delta x_n - a\Delta t_n}{b\sqrt{2\pi}\Delta t_n}\right)^2\right) \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(\frac{-a^2}{2\sigma^2}\right) = \\ &= \exp\left(\log\left(\prod_{n=1}^N \frac{1}{b\sqrt{2\pi}\Delta t_n} \exp\left(-\left(\frac{\Delta x_n - a\Delta t_n}{b\sqrt{2\pi}\Delta t_n}\right)^2\right)\right)\right) \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(\frac{-a^2}{2\sigma^2}\right) = \\ &= \exp\left(\sum_{n=1}^N \log\left(\frac{1}{b\sqrt{2\pi}\Delta t_n}\right) - \frac{(\Delta x_n - a\Delta t_n)^2}{2b^2\Delta t_n}\right) \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(\frac{-a^2}{2\sigma^2}\right) \propto \\ &\propto \exp\left(-\sum_{n=1}^N \frac{(\Delta x_n - a\Delta t_n)^2}{2b^2\Delta t_n} - \frac{a^2}{2\sigma^2}\right) = \\ &= \exp\left(-a^2\left(\sum_{n=1}^N \frac{\Delta t_n}{2b^2} + \frac{1}{2\sigma^2}\right) + a\left(\sum_{n=1}^N \frac{\Delta x_n}{b^2}\right) - \left(\sum_{n=1}^N \frac{\Delta x_n^2}{2b^2\Delta t_n}\right)\right) \propto \\ &\propto \exp\left(-a^2\left(\sum_{n=1}^N \frac{\Delta t_n}{2b^2} + \frac{1}{2\sigma^2}\right) + a\left(\sum_{n=1}^N \frac{\Delta x_n}{b^2}\right)\right) \end{aligned}$$

In order to complete the square term of the argument of the exponential function we can add and subtract the term:

$$\left(\frac{\sum_{n=1}^N \frac{\Delta x_n}{b^2}}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}}\right)^2$$

So we get:

$$\begin{aligned} &\exp\left(-a^2\left(\sum_{n=1}^N \frac{\Delta t_n}{2b^2} + \frac{1}{2\sigma^2}\right) + a\left(\sum_{n=1}^N \frac{\Delta x_n}{b^2}\right) - \left(\frac{\sum_{n=1}^N \frac{\Delta x_n}{b^2}}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}}\right)^2\right) \exp\left(\left(\frac{\sum_{n=1}^N \frac{\Delta x_n}{b^2}}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}}\right)^2\right) \propto \\ &\propto \exp\left(-a^2\left(\sum_{n=1}^N \frac{\Delta t_n}{2b^2} + \frac{1}{2\sigma^2}\right) + a\left(\sum_{n=1}^N \frac{\Delta x_n}{b^2}\right) - \left(\frac{\sum_{n=1}^N \frac{\Delta x_n}{b^2}}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}}\right)^2\right) = \end{aligned}$$

$$= \exp\left(\frac{-(a - \frac{\sum_{n=1}^N \frac{\Delta x_n}{b^2}}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}})^2}{2 \frac{1}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}}}\right)$$

At this point we can recognize the gaussian form and the expressions for mean value and variance (we substitute some terms with the given values $t_N = 1, t_0 = 0$):

$$\bar{\sigma} = \sqrt{\frac{1}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}}} = \frac{1}{\sqrt{\frac{1}{\sigma^2} + \frac{1}{b^2}}}$$

$$\mu(x, b) = \frac{\sum_{n=1}^N \frac{\Delta x_n}{b^2}}{\sum_{n=1}^N \frac{\Delta t_n}{b^2} + \frac{1}{\sigma^2}} = \frac{\bar{\sigma}(x, b)^2}{b^2}(x_N - x_0)$$

2.2

As done before, we start by applying Bayes' rule:

$$p(B^2|X, A) \propto p(X|A, B^2)p(B^2)$$

proof is done in a similar way as the one for previous subsection. Here we want to write $p(B^2|X, A)$ in a way such that we can recognize the general form of an inverse gamma distribution:

$$p(x, \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\beta}{x}\right) \propto x^{-\alpha-1} \exp\left(-\frac{\beta}{x}\right)$$

Here we start with the computations, first decomposing $p(x|a, b^2)$:

$$p(x|a, b^2) = \prod_{n=1}^N \frac{1}{\sqrt{2\pi b^2 \Delta t_n}} \exp\left(-\frac{(\Delta x_n - a \Delta t_n)^2}{2b^2 \Delta t_n}\right) \propto$$

$$\propto (b^2)^{-\frac{N}{2}} \exp\left(-\frac{1}{2b^2} \sum_{n=1}^N \frac{(\Delta x_n - a \Delta t_n)^2}{\Delta t_n}\right)$$

Now, remembering that:

$$p(b^2) = \frac{\beta^\alpha}{\Gamma(\alpha)} (b^2)^{-(\alpha+1)} \exp\left(-\frac{\beta}{b^2}\right)$$

We can combine these two expressions doing their product:

$$p(b^2|x, a) = p(x|a, b^2)p(b^2) \propto (b^2)^{-\frac{N}{2}} \exp\left(-\frac{1}{2b^2} \sum_{n=1}^N \frac{(\Delta x_n - a \Delta t_n)^2}{\Delta t_n}\right) \frac{\beta^\alpha}{\Gamma(\alpha)} (b^2)^{-(\alpha+1)} \exp\left(-\frac{\beta}{b^2}\right) \propto$$

$$\propto (b^2)^{-\frac{N}{2}-\alpha-1} \exp\left(-\frac{1}{b^2} \left(\beta + \sum_{n=1}^N \frac{(\Delta x_n - a \Delta t_n)^2}{2\Delta t_n}\right)\right)$$

At this point we can see the classic form of an inverse Gamma distribution, with parameters:

$$\bar{\alpha}(x, a) = \frac{N}{2} + \alpha$$

$$\bar{\beta}(x, a) = \beta + \sum_{n=1}^N \frac{(\Delta x_n - a \Delta t_n)^2}{2\Delta t_n}$$

2.3

See code in the attached file

2.4

Convergence of Gibbs Samples

Here I put the three 2D histograms for the values of the parameters a and b obtained through Gibbs sampling with respectively values of $K_{\max} = 200, 2000, 20000$, in order to show the convergence of the samples to a stationary distribution.

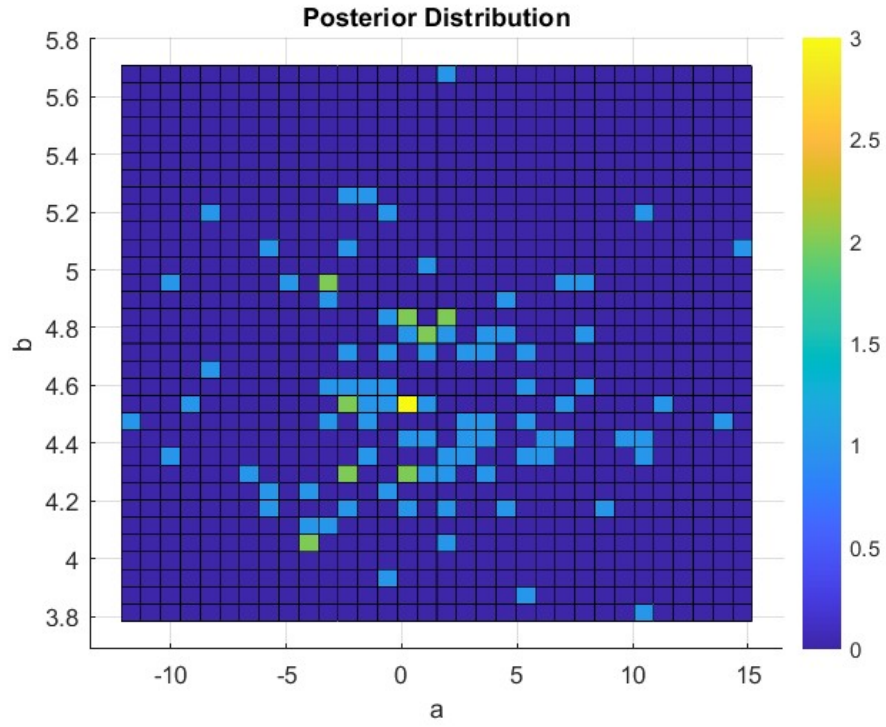


Figure 2: $K_{\max} = 200$

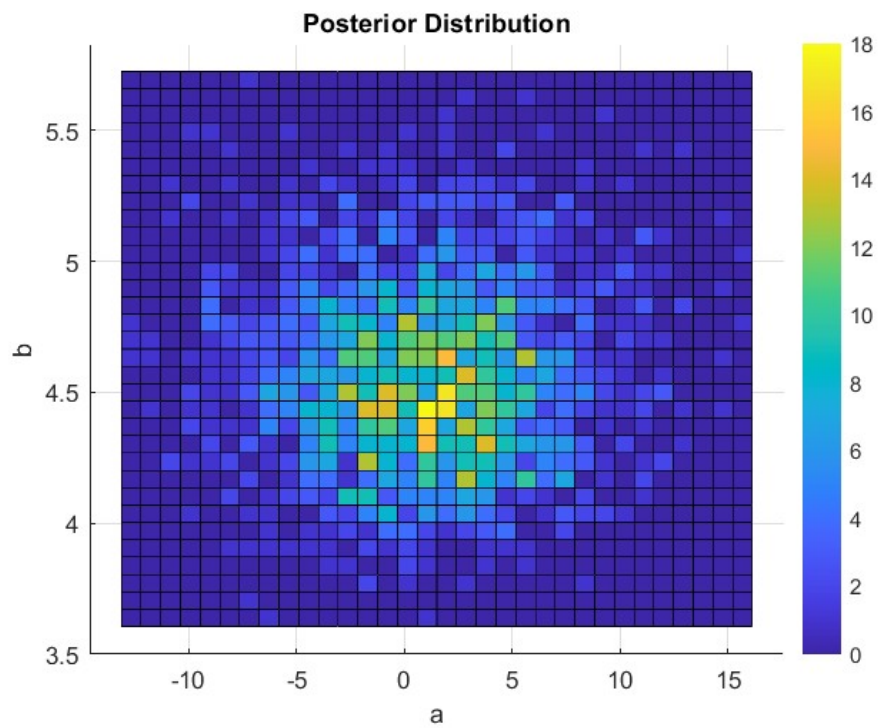


Figure 3: $K_{max} = 2000$

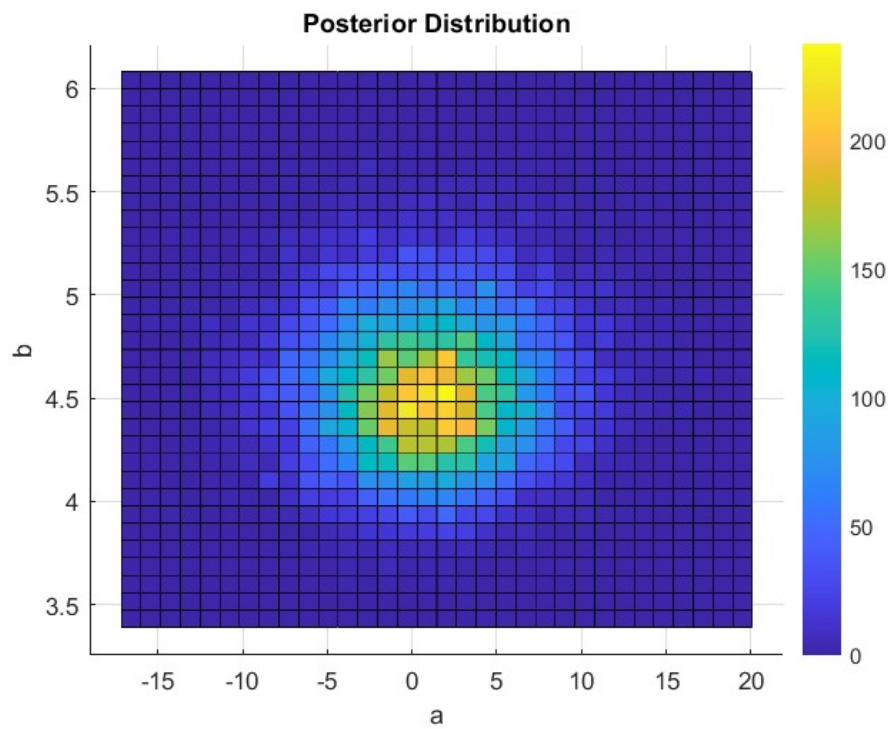


Figure 4: $K_{max} = 20000$

2.5

After averaging over the Gibbs iterates, for different values of K_{max} , I obtained values of the parameters around the following:

$$a_{post} = 0.9873 \quad b_{post} = 0.2103$$

While the values for the maximizer of the likelihood function are:

$$a_{ML} = 0.9515 \quad b_{ML} = 0.9900$$

2.6

Here I got 500 Gibbs samples (500 couples of a, b), each attained after 1000 steps of the Gibbs method. Then using the formula that we derived before (subsection 1.3) for $X_{N+1}|X$:

$$(X_{N+1}|X) \sim \mathcal{N}(x_N + a\Delta t_{N+1}, b^2\Delta t_{N+1})$$

I got 500 values of X_{N+1} and plotted them in a histogram, to show the gaussian pattern:

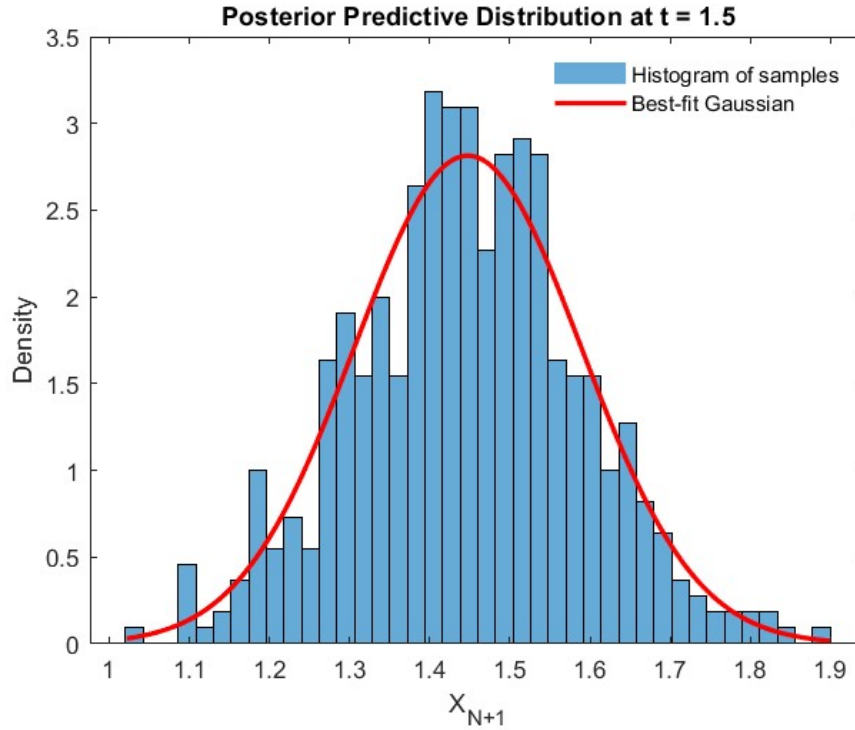


Figure 5: X_{N+1} with Gibbs sampling

Why does this produce a sample from the posterior predictive distribution $p(X_{N+1}|X)$?
Because from the law of total probability follows that:

$$p(X_{N+1}|x) = \int \int p(X_{N+1}|x, a, b)p(a, b|x)$$

and what we are doing here is approximating these integrals sampling (a_k, b_k) from the posterior $p(a, b|x)$ and then, as written above, sampling $X_{N+1}^{(k)} \sim p(X_{N+1}|x, a, b)$, so this method samples from the true predictive distribution $p(X_{N+1}|x)$ by combining the posterior samples of the parameters and the conditional distribution of X_{N+1} of those parameters.

Comparison

3.1

With the frequentist approach, we derived fixed values for the parameters a and b , choosing the ones that maximized the likelihood function (this procedure is based on the idea that "If a certain event happened, then it is probably happening relatively frequently"), then we predicted the distribution of X_{N+1} using these fixed values a_{ML}, b_{ML} :

$$(X_{N+1}|x) \sim \mathcal{N}(x_N + a_{ML}\Delta t_{N+1}, b_{ML}^2\Delta t_{N+1})$$

On the other hand, with the Bayesian approach, we considered the parameters a, b as random variables with a certain posterior distribution $p(a, b|x)$. We assumed that the samples derived from the Gibbs method were distributed following this posterior probability density function, so we could obtain a sample $X_{N+1}^{(k)}$ for each couple of samples a_k, b_k :

$$(X_{N+1}^{(k)}|X = x) \sim \mathcal{N}(x_N + a_k\Delta t_{N+1}, b_k^2\Delta t_{N+1})$$

3.2

As suggested, we modified the values of σ, α and β increasing σ and β and decreasing α in order to check the effects of an uninformative prior on the prediction of values of X_{N+1} . The distribution obtained this time for X_{N+1} is represented by the following histogram:

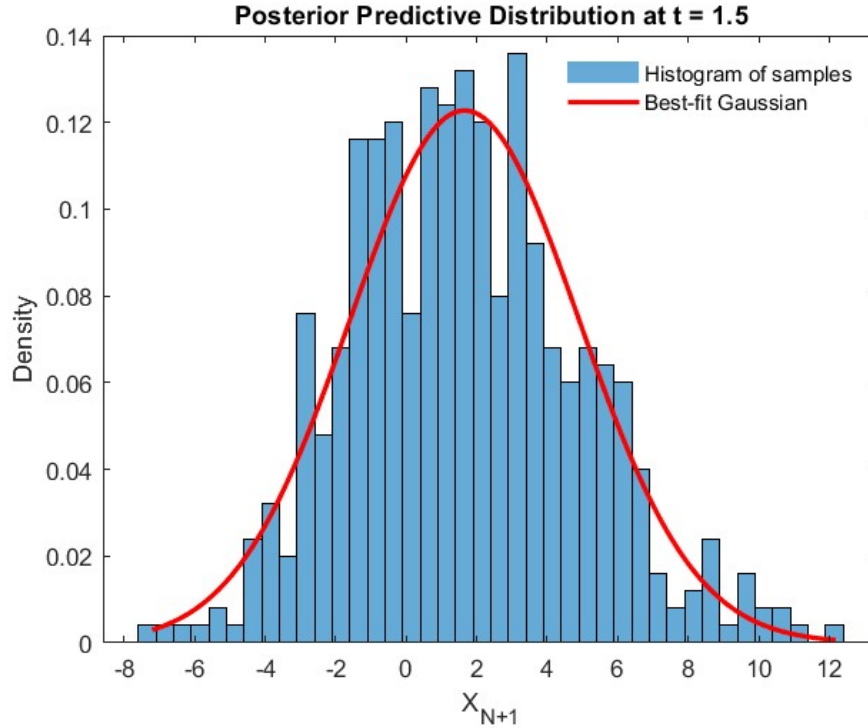


Figure 6: X_{N+1} with uninformative prior

It is important here to look how spread is the distribution over the domain. In fact, when we predicted the probability density of X_{N+1} with the given values in 2.6, the function was a gaussian, spreading, mostly, over an interval of length 0.9 (interval $[1; 1.9]$), so we could have a quite precise approximation for X_{N+1} choosing one of the values in this short interval. When it comes to generate the prediction using an uninformative prior (in the histogram I used values $\sigma = 1000, \alpha = 0.1, \beta = 1000$), the gaussian function is way more flat

and spread on an interval of length 20 (interval $[-8; 12]$), making it more difficult to pick a proper value for X_{N+1} .

3.3

One of the main advantages of the Bayesian estimation approach is the fact that it explicitly models the uncertainty over the parameters, while the frequentist approach does not account for the variance of the parameters' values, it only gives a fixed esteem. Moreover, the Bayesian approach allows us to incorporate our prior beliefs about the parameters (this aspect can also be seen as a drawback of the Bayesian version, since in order to obtain a good result, we need some kind of prior knowledge about the parameters, which is not always available and it has to be informative and precise, otherwise we would not get a reliable result, as seen in the previous subsection).

On the other hand, the frequentist approach has a simpler interpretation, it does not require any prior knowledge about parameters and it is faster than the Bayesian one, in fact we only need to find the minimizer of the $-\log(\text{likelihood})$ function, which usually only implies computing the gradient of the function and solve a system of equations, while for the Bayesian version we have to run a Monte Carlo simulation in order to obtain the parameters' samples).

With this information, we can say that the Bayesian estimation approach generally produces better and more informative predictions, but the frequentist approach is simpler and has a faster implementation.

3.4

Extending to SDEs with non constant parameters, for example with non constant drift terms:

$$dX_t = f(X_t, t)dt + b dW_t$$

the solution often does not have a closed form, so we have to approximate it using numerical methods (e.g. forward Euler). Then, since the drift is no more constant, writing the expression for the likelihood function becomes difficult (we cannot use anymore the linear transformations' properties that we used earlier).